

1. A **square** (respectively **cubic**) integer is an integer that factors as a^2 (respectively a^3) for $a \in \mathbb{Z}$.

1. Prove that when a square integer is divided by 3, the remainder can never be 2.

2. Prove that a cubic integer can be written in the form $9k$, $9k+1$, or $9k-1$ for some $k \in \mathbb{Z}$.

Pf. 1. We prove by cases.

By Corollary 2.5, $\mathbb{Z}$ is separated by $[1]_3$, $[2]_3$ and $[0]_3$ exactly,

So any integer must be either $3k$, $3k+1$ or $3k+2$ for some integer k .

Let a be an arbitrary integer

So a^2 is an arbitrary square integer.

Case 1 $a = 3k$ for some $k \in \mathbb{Z}$

Then $a^2 = 9k^2 = 3(3k^2)$, $3|a^2$.

Case 2 $a = 3k+1$ for some $k \in \mathbb{Z}$

Then $a^2 = (3k+1)^2 = 3(3k^2+2k)+1$
the remainder when a^2 divided
by 3 is 1.

Case 3 $a = 3k+2$ for some $k \in \mathbb{Z}$.

Then $a^2 = (3k+2)^2 = 3(3k^2+4k)+1$
the remainder when a^2 divided
by 3 is 1.

Since the 3 cases cover all situations and in none of them the remainder is 2, we have completed the proof.

2. Let a be an arbitrary integer.
Then a^3 is an arbitrary cubic integer.
Same as part I, we divide all situations into 3 cases:

Case 1 $a = 3m$ for some integer m .

$$\text{Then } a^3 = 27m^3 = 9(3m^3)$$

$$\text{So take } k = 3m^3, a^3 = 9k$$

Case 2 $a = 3m + 1$ for some integer m

$$\text{Then } a^3 = 27m^3 + 27m^2 + 9m + 1$$

$$\text{So take } k = 3m^3 + 3m^2 + m, a^3 = 9k + 1$$

Case 3 $a = 3m + 2$ for some integer m .

$$\text{Then } a^3 = 27m^3 + 54m^2 + 36m + 8$$

$$\text{So take } k = 3m^3 + 6m^2 + 4m + 1, \\ a^3 = 9k - 1$$

Since this covers all situations,
we have proved the statement.

2. Analogous to the idea of "greatest common divisor" is the idea of "least common multiple." The least common multiple of two (positive) integers, a and b , is the smallest (positive) integer m such that $a|m$ and $b|m$.
- Use the well-ordering principle to prove that the least common multiple of two integers always exists.
 - Let $d = \gcd(a, b)$. Prove that there is some integer m such that $dm = ab$.
 - Prove that m from part (b) is divisible by a and divisible by b .
 - Prove that for all integers M such that M is a common multiple of a and b , the number $m = \frac{ab}{\gcd(a, b)}$ (from parts b and c) divides M , making m the least common multiple.
(Hint: prove that $\frac{M}{m}$ is an integer using Bezout's identity).

(a) Pf Let a, b be arbitrary positive integers.

$$\text{Let } S = \{s \in \mathbb{Z}^+ \mid a|s \text{ and } b|s\}$$

First, $ab \in S$, so S is non-empty

Also, since any element $s \in S$ is also an element of $\mathbb{Z}^+$, $S \subseteq \mathbb{Z}^+ \subseteq \mathbb{Z}_{\geq 0}$

So by well-ordering principle, S has a smallest element, then by definition, the least common multiple of a, b exists.

Since a, b is arbitrary, we have proved the statement.

(b) Pf Since $d|a, d|b$ by definition of $\gcd$
 $a = dp, b = dq$ for some $p, q \in \mathbb{Z}$

So $ab = dpq$. Take pq as m , then $dm = ab$.

(c) Since $dm = ab$, $a = dp$

we have $dm = dpb$

$$m = pb \text{ where } p \in \mathbb{Z} \text{ in (b)}$$

Similarly, we have $m = qa$ where $q \in \mathbb{Z}$ in (b)

So $a|m$, $b|m$.

(d) Let M be an arbitrary common multiple of a, b . Then $M = sa = kb$ for some integers s, k .

By Bezout, $(a, b) = ra + sb$ for some integers r, s .

$$\text{So } m = \frac{ab}{(a, b)} = \frac{ab}{ra + sb}, \quad \frac{M}{m} = \frac{ra + sb}{\frac{ab}{m}} \cdot kb \\ = rk + \frac{ksb}{a}$$

$$\text{Since } kb = sa, \quad \frac{M}{m} = rk + \frac{s^2a}{a} = rk + s^2$$

Since $r, k, s \in \mathbb{Z}$, $\frac{M}{m}$ is an integer
so $m|M$.

And this implies for all multiple of a, b

$$M, \quad m \leq M. \text{ Also, } m = \frac{ab}{(a, b)} = k_1 b = k_2 a$$

for some integers k_1, k_2 since $(a, b) | a$,

$(a, b) | b$. Therefore m is a common multiple of a, b , and is the least common multiple.

3. Let $a, b, c, d \in \mathbb{Z}$ such that the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

has an inverse whose entries are all integers. Let $x, y \in \mathbb{Z}$ such that at least one of them is not zero. Prove that $\gcd(x, y) = \gcd(ax + by, cx + dy)$.¹

¹How can we relate x and y with $ax + by$ and $cx + dy$ using the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$?

Denote $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ by A and its inverse by A^{-1} .

Note that $\det A = ad - bc \in \mathbb{Z}$, and $\det A^{-1}$

Therefore $\det A = ad - bc = 1$ or -1 , $\det A^{-1} = \frac{1}{\det A} \in \mathbb{Z}$.

$$\text{So } A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (\pm 1) \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\text{Then } A^{-1} \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} = \begin{pmatrix} (ad-bc)x \\ (ad-bc)y \end{pmatrix} = (\pm 1) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{So } x = (\pm 1) [d(ax+by) - b(cx+dy)]$$

$$y = (\pm 1) [-c(ax+by) + a(cx+dy)]$$

This implies that x and y are both

linear combinations of $(ax+by)$ and $(cx+dy)$

So any common divisor of $(ax+by)$ and $(cx+dy)$ is also common divisor of x and y

So $\gcd(ax+by, cx+dy)$ is a common divisor of x, y . ①

And since $ax+by, cx+dy$ are linear combinations of x, y . By the same logic, $\gcd(x, y)$ is a common divisor of $ax+by$ and $cx+dy$. ②

By ①, ②, $\gcd(x, y) = \gcd(ax+by, cx+dy)$.

To show this, denote $ax+by$ as m and $cx+dy$ as n . We know $\gcd(m,n) \mid x$, $\gcd(m,n) \mid y$, so $\gcd(m,n) \leq \gcd(x,y)$

Same we have $\gcd(x,y) \leq \gcd(m,n)$

So $\gcd(m,n) = \gcd(x,y)$.

Then we have finished the proof.

4. For any integer m , we can use the Fundamental Theorem of Arithmetic to write $m = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}$ where the p_i 's are distinct primes in an (essentially) unique way. The natural number a_i is said to be the multiplicity of the prime p_i in m . [By convention, the multiplicity of p in m is 0 if p does not divide m .]

1. Let d and n be positive integers. Prove that n is a d -th power of some other integer if and only if for every prime p , the multiplicity of p in n is divisible by d .
2. Prove that if n is not a d -th power of some other integer, then $\sqrt[d]{n}$ is irrational. [Hint: try proof by contradiction.]

1. Proof Note the two statements as ①, ② respectively.
First, we prove ① $\rightarrow$ ②

(1) Let $n = \beta^d$ for some integer β .

Let p be an arbitrary prime number.

We want to show that

for some integer k , $p^k \mid n$ and $d \mid k$.

By FTA we know $\beta = q_1^{a_1} q_2^{a_2} \dots q_m^{a_m}$ for
some primes $q_1, \dots, q_m$ and integers
 $a_1, a_2, \dots, a_m$

$$\text{Then } n = q_1^{a_1} q_2^{a_2} \dots q_m^{a_m}$$

We prove the statement by cases.

Case 1 $p = q_i$ for some $1 \leq i \leq m$.

Then consider $k = a_i d$

$$\text{We know } n = q_i^{a_i} \cdot (q_1^{a_1} \dots q_{i-1}^{a_{i-1}} \cdot q_{i+1}^{a_{i+1}} \dots q_m^{a_m})$$

$$\text{So } q_i^{a_i d} \mid n, \text{ that is, } \underline{p^k \mid n}$$

Also, since $k = a_i d$, $\underline{d \mid k}$

This completes the proof of $\textcircled{1} \rightarrow \textcircled{2}$

Next we prove $\textcircled{2} \rightarrow \textcircled{1}$

(2) Let p be an arbitrary prime, n be
an arbitrary integer with the property
that for any prime p , the multiplicity of
 p in n is some integer a s.t. $a = kd$ for
some integer k .

By FTA, we know $n = p_1^{a_1} p_2^{a_2} \dots p_m^{a_m}$

for some primes $p_1, p_2, \dots, p_m$ and their multiplicity $a_1, a_2, \dots, a_m$

And by the property above we have

$a_1 = k_1 d, a_2 = k_2 d, \dots, a_m = k_m d$ for some integers $k_1, k_2, \dots, k_m$

Then $n = (p_1^{k_1} p_2^{k_2} \dots p_m^{k_m})^d$.

Thus there is always an integer

$(p_1^{k_1} p_2^{k_2} \dots p_m^{k_m})$ such that n is its d -th power.

This completes the proof of $(2) \rightarrow (1)$

Since we have $(1) \rightarrow (2)$, $(2) \rightarrow (1)$, we have proved the if-and-only-if relation between the two statements.

2 - Proof We prove it by contradiction.

Let n be an arbitrary integer s.t. n is not a d -th power of any integer,

Assume for contradiction that $\sqrt[d]{n}$ is a $d \in \mathbb{Z}^+$ rational number. So $\sqrt[d]{n} = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$, where $(p, q) = 1$.

Then $n = \frac{p}{q^d}$, $nq^d = p^d$.

By FTA, $q = q_1 q_2 \dots q_m$ for some primes

$$p = p_1 p_2 \dots p_s \text{ for some primes } q_1, \dots, q_m$$

Then $n (q_1^d q_2^d \dots q_m^d) = p_1^d p_2^d \dots p_s^d$

Since $(p, q) = 1$, none of $p_1, \dots, p_s$ is equal to any of $q_1, \dots, q_m$.

So n must contain the factor $p_1^d p_2^d \dots p_s^d$.

And since $n(q_1^d q_2^d \dots q_m^d) = p_1^d p_2^d \dots p_s^d$,
where $p_1, \dots, p_s$ are all primes, any prime factor of n must be among $p_1, p_2, \dots, p_s$.

So all prime factors of n have the multiplicity of d , then by part I we know that n is the d^{th} power of some integer which contradicts the condition.

Then we have completed the proof.

5. **We will see many equivalence relations in this course, and we will apply this problem many times during the semester.** Let X be a non-empty set, and let $\sim$ be an equivalence relation on the set X , that is, a rule used to compare pairs of elements of X which satisfies the following properties (we read “ $a \sim b$ ” as “ a is related to b ”):

- $\sim$ is reflexive: $a \sim a$ for all $a \in X$.
- $\sim$ is symmetric: let $a, b \in X$ such that $a \sim b$. Then $b \sim a$.
- $\sim$ is transitive: let $a, b, c \in X$ such that $a \sim b$ and $b \sim c$. Then, $a \sim c$.

For example, before you start this problem note that the relation $=$ (being equal to) is an equivalence relation on any non-empty set X . However, the relation $\leq$ (being less or equal to) on the set $\mathbb{R}$ is not an equivalence relation, as it is not symmetric. Determine whether the following rules are equivalence relations on the given sets.

(a) Let X be $\mathbb{R}^2$, and $\sim$ is the rule

$$\begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} w \\ z \end{bmatrix}$$

if $x - y = w - z$.

(b) Let X be the set of integers, $\mathbb{Z}$, and let $\sim$ be the rule

$$a \sim b$$

if $ab \geq 0$.

(c) Given an equivalence relation $\sim$ in X , we define the **equivalence class of a** (denoted by $[a]$) for all $a \in X$ as

$$[a] := \{x \in X : x \sim a\} \subseteq X.$$

Show that, for all $a \in X$, $[a] \neq \emptyset$.

(d) Let $a, b \in X$. Show that either $[a] \cap [b] = \emptyset$ or $[a] = [b]$.² That is, prove that **equivalence classes are either disjoint or equal**.

(e) Let $\mathcal{S} := \{[a] : a \in X\}$ the set of all equivalence classes with respect to the relation $\sim$. Prove that

$$X = \bigsqcup_{Y \in \mathcal{S}} Y,$$

(the symbol $\sqcup$ means *disjoint union*). In other words, prove that **equivalence classes partition X into disjoint sets**.

²Hint: Suppose that $[a] \cap [b] \neq \emptyset$, that is, there exists $c \in X$ such that $c \in [a] \cap [b]$. Do you remember how to prove “or” statements?

(a) Let $\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} w \\ z \end{bmatrix} \in \mathbb{R}^2$ be arbitrary, with $x-y = w-z$

Since $x-y = x-y$,

$\begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} x \\ y \end{bmatrix}$, meeting reflective.

Since $x-y = w-z$, $w-z = x-y$

This implies $\begin{bmatrix} w \\ z \end{bmatrix} \sim \begin{bmatrix} x \\ y \end{bmatrix}$ whenever $\begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} w \\ z \end{bmatrix}$,
meeting symmetric

Let $\alpha, \beta \in \mathbb{R}$ be arbitrary with $w-z = \alpha - \beta$

Since $x-y = w-z$, $\begin{bmatrix} w \\ z \end{bmatrix} \sim \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

We have $x-y = \alpha - \beta \Leftrightarrow \begin{bmatrix} x \\ y \end{bmatrix} \sim \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

This implies transitive

So the rule is a equivalence relation

(b) For transitivity, consider $a=1$, $c=-1$, $b=0$

Then $ab=0$, so $a \sim b$

$bc=0$, so $b \sim c$

But $ac=-1 < 0$, $a \not\sim c$

In this case $a \sim b$, $b \sim c$ and $a \not\sim c$

So the rule does not meet transitive.

Therefore it is not an equivalence relation.

cc) Let a be an arbitrary element of X .

By reflexive property of equivalence relation,

$a \sim a$ and since we know $a \in X$,

a must be in $\{x \in X \mid x \sim a\}$

So the set contains at least one element,

Therefore for all $a \in X$, $[a] \neq \emptyset$ since our selection of a is arbitrary.

(d) Assume $[a] \cap [b] \neq \emptyset$

Then $\exists x \in X$ s.t. $x \in [a]$ AND $x \in [b]$

Let m be an arbitrary element of $[a]$, by

definition we have $m \sim a$

since $x \sim a$ AND $m \sim a$, by transitive we have
 $m \sim x$.
(also by symmetric)

Since the selection of m is arbitrary,

$$[a] = \{x \in X \mid x \sim a\}$$

Similarly, let n be an arbitrary element of $[b]$,

We will see $n \sim x$, so $[b] = \{x \in X \mid x \sim n\}$

So $[a] = [b]$

Therefore either $[a] \cap [b] = \emptyset$ or $[a] = [b]$.

(e) Let x be an arbitrary element of X .

By reflexive property of equivalence class,

$x \sim x$, and thus $[x]$ is non-empty since it at least contains x

So $[x] \in S$.

Therefore $\forall x \in X, [x] \in S$

Hence $X = \bigcup_{Y \in S} Y$

And by (d) we have proved that $\forall x, y \in X$

either $[x] = [y]$ or $[x] \cap [y] = \emptyset$

So by definition, $\forall [x], [y] \in Y$, they are disjoint.

So $X = \bigsqcup_{Y \in S} Y$ by definition.